



Degree	University/Institute	Year	CGPA/Marks(%)
M.Tech Computer Science & Engineering / NIS - TA	IIT Hyderabad	2026	7.60
B. Tech Computer Science & Engineering	Institute of Engineering and Technology (DAVV) Indore	2021	7.08
XII (CBSE)	Jawahar Navodaya Vidyalaya Khargone	2016	71.60%
X (CBSE)	Jawahar Navodaya Vidyalaya Khargone	2014	8.40

WORK EXPERIENCE

Software Engineer | Infosys – Pune | June 2022 – June 2024

- Worked on **HSBC's Juniper data system**, managing **real-time data processing pipelines** using **Apache NiFi** and **Apache Kafka** on **Linux**. Configured and optimized **data flows** to meet evolving business needs, ensuring high availability and low latency. Monitored, troubleshooted, and resolved integration issues via **Jira**, maintaining **reliable data ingestion** and transformation workflows.

Tools & Technologies: Apache NiFi, Apache Kafka, Linux, SQL, Shell Scripting, Jira, Confluence

- Worked on a **capstone project** to develop a **Spring Boot-based** Employee Management System enabling CRUD operations for employee records. Built **RESTful APIs** and integrated backend services with **MySQL** for efficient data handling, ensuring a clean, **microservices-based architecture** with **Git-based version control**.

Tools & Technologies: Java, Spring Boot, MySQL, REST APIs, Hibernate, Maven, Postman, Git

Intern — Marvell Technology | Hyderabad | June – Aug 2025

- Reviewed **MoE architectures** in **LLMs** (Large Language Models), analyzing routing paradigms (**Softmax Top-k**, **BASE**, **Domain-Aware**, **Expert-Choice**) and models like **Mixtral (8×7B)**. Evaluated efficiency–capacity **trade-offs** and routing stability, proposing depth-aware and context-adaptive frameworks for improved scalability.

Tools & Technologies: Hugging Face Transformers, LaTeX, Google Scholar, arXiv, Papers with Code, Jupyter Notebooks

PROJECTS

Capstone Project: AI for Waste Management | Computer Vision, Object Detection, Image Segmentation [\[Github\]](#)

- Developing an **AI-powered multi-platform waste detection system** for **MRFs** (Material Recovery Facilities), **UAVs** (Unmanned Aerial Vehicles), and **surveillance networks**, enabling **real-time** waste identification and **monitoring**.
- Implementing and benchmarking **YOLO**, **EfficientDet**, and **DETR** for **object detection**, and **SAM** and **DeepLabV3** for **image segmentation** to enhance accuracy and efficiency.
- Planning to integrate **foundation models** such as **CLIP**, **DINO**, and **Florence-2**, and design a **chatbot interface** for interactive AI-assisted waste analysis.

Sensor-Based Human Activity Classification | Python, Pandas, Scikit-learn, Matplotlib, NumPy. [\[Github\]](#)

- Developed a **CNN–LSTM model** on the PIRvision dataset for **human activity recognition**, achieving **~90% accuracy**.
- Improved data quality using 10+ engineered **statistical** and **temporal features**, **outlier detection**, **feature scaling**, and **class balancing** techniques.
- Compared and evaluated **MLP**, **CNN**, **GRU**, and **CNN–LSTM** models through **cross-validation**, achieving a **15% accuracy improvement** and **robust IoT-based monitoring performance**.

Personalized Federated Learning | Python, PyTorch, NumPy, Scikit-learn, Matplotlib, Git. [\[Github\]](#)

- Reproduced and enhanced the **FedSelect** algorithm for client-specific model **personalization** in federated learning.
- Applied **Lottery Ticket Hypothesis–based** and **gradient-driven subnetwork selection** for adaptive parameter tuning.
- Identified the **novel approach** proposed in the FedSelect paper and integrated **reinforcement learning** to dynamically optimize personalization parameters (prune rate and sparsity), enhancing model scalability and robustness.

Tunneling and Offload with Linux and VPP | Networking Protocols, VXLAN, VPP, DPU, IPsec, WireGuard [\[Blog\]](#)

- Explored **IPsec** and **WireGuard** tunneling protocols to understand secure and private network communication.
- Configured **VXLAN tunnels** between **Linux** and **DPU systems** to simulate multi-layer data transmission.
- Gained hands-on experience with **Linux** and **VPP**, demonstrating **network offload** and performance gains using **DPUs**.

SKILLS

Languages: Java, Python, SQL, JavaScript, HTML, CSS, Shell Scripting

Frameworks & Technologies: Spring Boot, Apache Kafka, Apache NiFi, Hibernate, PyTorch, TensorFlow, Scikit-learn, Hugging Face Transformers, Pandas, NumPy, Matplotlib, LaTeX

Machine Learning & Deep Learning: CNNs, RNNs, Transformers, Generative Models (GANs, VAEs, Diffusion Models)

Tools & Platforms: MySQL, MongoDB, Linux, REST APIs, Git/GitHub, Jira, Wireshark, VS Code, Google Colab, Kaggle

Problem Solving: Strong foundation in Data Structures & Algorithms; practiced on GeeksforGeeks, LeetCode, HackerRank

RELEVANT COURSES

Foundations of Machine Learning | Deep Learning | Advanced Computer Networks | Advanced Data Structures and Algorithms | Topics in Networks | Multimedia Content Analysis | Linear Optimization | Fraud Analytics

POSITION OF RESPONSIBILITY

As a Teaching Assistant for the DBMS course at IIT Hyderabad, I supported the faculty in delivering course content, guided students during tutorials, and assisted in evaluating assignments and projects.

EXTRACURRICULAR

Actively involved in cultural and sports activities, performed tabla at the cluster level and volleyball at the regional level.

CERTIFICATIONS

- Getting Started with Deep Learning – NVIDIA Deep Learning Institute (DLI)
- Introduction to Transformer-Based NLP – NVIDIA Deep Learning Institute (DLI)